MongoDB 2

1) Apartado A

Escribe los comandos necesarios para realizar las siguientes operaciones DML sobre la colección zips:

1) Crea una entrada con los siguientes datos:

```
{
        city: 'OVIEDO',
        zip: '33001',
        loc: {x: 43.3625025223589, y: -5.843830860844533},
        pop: 226872,
        state: 'España'
}
```

Lo añadimos.

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.insert({city: 'OVIEDO',zip: '33001',loc: {x: 43.3625025223589, y: -5.843830860844533},pop: 226872,state: 'Esp
aña'})
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('699d5949dd4a3d6c22628ca0') }
}
```

2) Crea una entrada con los datos del código postal donde vives (si es el mismo código postal, crea uno diferente), con el valor de state a España.

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.insert({city: 'MIERES',zip: '33600',loc: {x: 43.1507025223589, y: 5.463530860844533},pop: 226872,state: 'Espa
ña'})
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('699d59cedd4a3d6c22628ca1') }
}
```

3) Modifica la población de tu código postal a 1.000.000.

```
db.zips.updateOne({zip: '33600'}, {$set: {pop: 1000000}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.updateOne({zip: '33600'}, {$set: {pop: 1000000}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({zip: '33600'})
[
  {
    _id: ObjectId('699d59cedd4a3d6c22628ca1'),
    city: 'MIERES',
    zip: '33600',
    loc: { x: 43.1507025223589, y: 5.463530860844533 },
    pop: 1000000,
    state: 'España'
  }
]
```

4) Incrementa la población de todos los documentos de España en 666 personas.

```
db.zips.updateMany({state: 'España'}, {$inc: {pop: 666}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.updateMany({state: 'España'}, {$inc: {pop: 666}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({state: 'España'})
[
  {
    _id: ObjectId('699d5949dd4a3d6c22628ca0'),
    city: 'OVIEDO',
    zip: '33001',
    loc: { x: 43.3625025223589, y: -5.843830860844533 },
    pop: 228870,
    state: 'España'
  },
  {
    _id: ObjectId('699d59cedd4a3d6c22628ca1'),
    city: 'MIERES',
    zip: '33600',
    loc: { x: 43.1507025223589, y: 5.463530860844533 },
    pop: 1000666,
    state: 'España'
  }
]
```

5) Añade un campo prov a ambos documentos con valor Alicante.

```
db.zips.updateMany({state: 'España'}, {$set: {prov: 'Asturias'}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.updateMany({state: 'España'}, {$set: {prov: 'Asturias'}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({state: 'España'})
[
  {
    _id: ObjectId('699d5949dd4a3d6c22628ca0'),
    city: 'OVIEDO',
    zip: '33001',
    loc: { x: 43.3625025223589, y: -5.843830860844533 },
    pop: 228870,
    state: 'España',
    prov: 'Asturias'
  },
  {
    _id: ObjectId('699d59cedd4a3d6c22628ca1'),
    city: 'MIERES',
    zip: '33600',
    loc: { x: 43.1507025223589, y: 5.463530860844533 },
    pop: 1000666,
    state: 'España',
    prov: 'Asturias'
  }
]
```

6) Renombra en los documentos de la provincia de Alicante el atributo prov por provincia

```
db.zips.updateMany({state: 'España'}, {$rename: {'prov': 'provincia'}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.updateMany({state: 'España'}, {$rename: {'prov': 'provincia'}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({state: 'España'})
[
  {
    _id: ObjectId('699d5949dd4a3d6c22628ca0'),
    city: 'OVIEDO',
    zip: '33001',
    loc: { x: 43.3625025223589, y: -5.843830860844533 },
    pop: 228870,
    state: 'España',
    provincia: 'Asturias'
  },
  {
    _id: ObjectId('699d59cedd4a3d6c22628ca1'),
    city: 'MIERES',
    zip: '33600',
    loc: { x: 43.1507025223589, y: 5.463530860844533 },
    pop: 1000666,
    state: 'España',
    provincia: 'Asturias'
  }
]
```

7) Elimina las coordenadas del zip 03206.

```
db.zips.updateMany({zip: '03206'}, {$unset: {'loc': ''}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.updateMany({zip: '03206'}, {$unset: {'loc': ''}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
```

8) Elimina tu entrada.

```
db.zips.deleteOne({zip: '33600'})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.deleteOne({zip: '33600'})
{ acknowledged: true, deletedCount: 1 }
```

2) Apartado B

Escribe los comandos necesarios para realizar las siguientes operaciones sobre la colección sample_restaurants.restaurants:

1) Insertar un nuevo restaurante de comida española con tu nombre, pero sin ninguna calificación.

```
db.restaurants.insert({address: {building: '1',coord: [ 34.00528899999999,
5.628886 ],street: 'Requejo',zipcode: '33600'},borough: 'Requejo',cuisine:
'Spanish', name: 'Pablo Menéndez',restaurant_id: '12345678'})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants> db.restaurants.insertOne({address: {building: '1',coord: [ 34.00528899999999, 5.628886 ],street: 'Requejo',zipcode
: '33600'},borough: 'Requejo',cuisine: 'Spanish', name: 'Pablo Menéndez',restaurant_id: '12345678'})
{
  acknowledged: true,
  insertedId: ObjectId('699ffafcebf2fd6e4a628ca3')
}
```

2) Añade una calificación al restaurante.

```
db.restaurants.updateOne({name: 'Pablo Menéndez'}, {$set: {grades: [{date:
new Date(), grade: 'A', score: 12 }]}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants> db.restaurants.updateOne({name: 'Pablo Menéndez'}, {$set: {grades: [{date: new Date(), grade: 'A', score: 12 }]}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants>  db.restaurants.find({name: 'Pablo Menéndez'})
[
  {
    _id: ObjectId('699d6220dd4a3d6c22628ca4'),
    name: 'Pablo Menéndez',
    grades: [
      { date: ISODate('2026-02-26T07:51:35.924Z'), grade: 'A', score: 12 }
    ]
  },
  {
    _id: ObjectId('699ffafcebf2fd6e4a628ca3'),
    address: {
      building: '1',
      coord: [ 34.00528899999999, 5.628886 ],
      street: 'Requejo',
      zipcode: '33600'
    },
    borough: 'Requejo',
    cuisine: 'Spanish',
    name: 'Pablo Menéndez',
    restaurant_id: '12345678'
  }
]
```

3) Cambiar el nombre del restaurante por Severo manjar.

```
db.restaurants.updateOne({name: 'Pablo Menéndez'}, {$set: {name: "Severo Manjar"}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants> db.restaurants.updateOne({name: 'Pablo Menéndez'}, {$set: {name: "Severo Manjar"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants>  db.restaurants.find({restaurant_id: '12345678'})
[
  {
    _id: ObjectId('699ffafcebf2fd6e4a628ca3'),
    address: {
      building: '1',
      coord: [ 34.00528899999999, 5.628886 ],
      street: 'Requejo',
      zipcode: '33600'
    },
    borough: 'Requejo',
    cuisine: 'Spanish',
    name: 'Severo Manjar',
    restaurant_id: '12345678',
    grades: [
      { date: ISODate('2026-02-26T07:55:16.006Z'), grade: 'A', score: 12 }
    ]
  }
]
```

4) Modifica el número del edificio de la dirección.

```
db.restaurants.updateOne({"address.street": 'Requejo'}, {$set: {"address.building": '2'}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants> db.restaurants.updateOne({"address.street": 'Requejo'}, {$set: {"address.building": '2'}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants>  db.restaurants.find({restaurant_id: '12345678'})
[
  {
    _id: ObjectId('699ffafcebf2fd6e4a628ca3'),
    address: {
      building: '2',
      coord: [ 34.00528899999999, 5.628886 ],
      street: 'Requejo',
      zipcode: '33600'
    },
    borough: 'Requejo',
    cuisine: 'Spanish',
    name: 'Severo Manjar',
    restaurant_id: '12345678',
    grades: [
      { date: ISODate('2026-02-26T07:55:16.006Z'), grade: 'A', score: 12 }
    ]
  }
]
```

5) Modifica el tipo de cocina para que sea Mediterránea.

```
db.restaurants.updateOne({restaurant_id: '12345678'}, {$set: {cuisine: 'Mediterránea'}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants> db.restaurants.updateOne({restaurant_id: '12345678'}, {$set: {cuisine: 'Mediterránea'}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants>  db.restaurants.find({restaurant_id: '12345678'})
[
  {
    _id: ObjectId('699ffafcebf2fd6e4a628ca3'),
    address: {
      building: '2',
      coord: [ 34.00528899999999, 5.628886 ],
      street: 'Requejo',
      zipcode: '33600'
    },
    borough: 'Requejo',
    cuisine: 'Mediterránea',
    name: 'Severo Manjar',
    restaurant_id: '12345678',
    grades: [
      { date: ISODate('2026-02-26T07:55:16.006Z'), grade: 'A', score: 12 }
    ]
  }
]
```

6) Añade una segunda calificación.

```
db.restaurants.updateOne({restaurant_id: '12345678'}, {$push: {grades: {date:
new Date(), grade: 'A', score: 15 }}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants> db.restaurants.updateOne({restaurant_id: '12345678'}, {$push: {grades: {date: new Date(), grade: 'A', score: 15 }}
})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants>  db.restaurants.find({restaurant_id: '12345678'})
[
  {
    _id: ObjectId('699ffafcebf2fd6e4a628ca3'),
    address: {
      building: '2',
      coord: [ 34.00528899999999, 5.628886 ],
      street: 'Requejo',
      zipcode: '33600'
    },
    borough: 'Requejo',
    cuisine: 'Mediterránea',
    name: 'Severo Manjar',
    restaurant_id: '12345678',
    grades: [
      { date: ISODate('2026-02-26T07:55:16.006Z'), grade: 'A', score: 12 },
      { date: ISODate('2026-02-26T08:27:15.339Z'), grade: 'A', score: 15 }
    ]
  }
]
```

7) Modifica las coordenadas de la ubicación para que coincidan con las de tu ciudad.

```
db.restaurants.updateOne({"address.street": 'Requejo'}, {$set:
{"address.coord": [34, 5]}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants> db.restaurants.updateOne({"address.street": 'Requejo'}, {$set: {"address.coord": [34, 5]}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants>  db.restaurants.find({restaurant_id: '12345678'})
[
  {
    _id: ObjectId('699ffafcebf2fd6e4a628ca3'),
    address: {
      building: '2',
      coord: [ 34, 5 ],
      street: 'Requejo',
      zipcode: '33600'
    },
    borough: 'Requejo',
    cuisine: 'Mediterránea',
    name: 'Severo Manjar',
    restaurant_id: '12345678',
    grades: [
      { date: ISODate('2026-02-26T07:55:16.006Z'), grade: 'A', score: 12 },
      { date: ISODate('2026-02-26T08:27:15.339Z'), grade: 'A', score: 15 }
    ]
  }
]
```

8) Modifica el barrio con el barrio donde vives.

```
db.restaurants.updateOne({restaurant_id: '12345678'}, {$set: {borough:
"Oñon"}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants>  db.restaurants.updateOne({restaurant_id: '12345678'}, {$set: {borough: "Oñon"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants>  db.restaurants.find({restaurant_id: '12345678'})
[
  {
    _id: ObjectId('699ffafcebf2fd6e4a628ca3'),
    address: {
      building: '2',
      coord: [ 34, 5 ],
      street: 'Requejo',
      zipcode: '33600'
    },
    borough: 'Oñon',
    cuisine: 'Mediterránea',
    name: 'Severo Manjar',
    restaurant_id: '12345678',
    grades: [
      { date: ISODate('2026-02-26T07:55:16.006Z'), grade: 'A', score: 12 },
      { date: ISODate('2026-02-26T08:27:15.339Z'), grade: 'A', score: 15 }
    ]
  }
]
```

9) Añade una tercera calificación que sea excelente.

```
db.restaurants.updateOne({restaurant_id: '12345678'}, {$push: {grades: {date:
new Date(), grade: 'A', score: 10000000 }}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants> db.restaurants.updateOne({restaurant_id: '12345678'}, {$push: {grades: {date: new Date(), grade: 'A', score: 10000
000 }}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants>  db.restaurants.find({restaurant_id: '12345678'})
[
  {
    _id: ObjectId('699ffafcebf2fd6e4a628ca3'),
    address: {
      building: '2',
      coord: [ 34, 5 ],
      street: 'Requejo',
      zipcode: '33600'
    },
    borough: 'Oñon',
    cuisine: 'Mediterránea',
    name: 'Severo Manjar',
    restaurant_id: '12345678',
    grades: [
      { date: ISODate('2026-02-26T07:55:16.006Z'), grade: 'A', score: 12 },
      { date: ISODate('2026-02-26T08:27:15.339Z'), grade: 'A', score: 15 },
      {
        date: ISODate('2026-02-26T08:33:21.945Z'),
        grade: 'A',
        score: 10000000
      }
    ]
  }
]
```

10) Modifica la calle y el código postal mediante una única operación.

```
db.restaurants.updateOne({"address.street": 'Requejo'}, {$set:
{"address.street": 'Oñon', "address.zipcode": '33610'}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants> db.restaurants.updateOne({"address.street": 'Requejo'}, {$set: {"address.street": 'Oñon', "address.zipcode": '3361
0'}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants>  db.restaurants.find({restaurant_id: '12345678'})
[
  {
    _id: ObjectId('699ffafcebf2fd6e4a628ca3'),
    address: { building: '2', coord: [ 34, 5 ], street: 'Oñon', zipcode: '33610' },
    borough: 'Oñon',
    cuisine: 'Mediterránea',
    name: 'Severo Manjar',
    restaurant_id: '12345678',
    grades: [
      { date: ISODate('2026-02-26T07:55:16.006Z'), grade: 'A', score: 12 },
      { date: ISODate('2026-02-26T08:27:15.339Z'), grade: 'A', score: 15 },
      {
        date: ISODate('2026-02-26T08:33:21.945Z'),
        grade: 'A',
        score: 10000000
      }
    ]
  }
]
```

11) Elimina la calificación más antigua.

```
db.restaurants.updateOne({restaurant_id: '12345678'}, {$pop:{"grades" : -1}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants>  db.restaurants.updateOne({restaurant_id: '12345678'}, {$pop:{"grades" : -1}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants>  db.restaurants.find({restaurant_id: '12345678'})
[
  {
    _id: ObjectId('699ffafcebf2fd6e4a628ca3'),
    address: { building: '2', coord: [ 34, 5 ], street: 'Oñon', zipcode: '33610' },
    borough: 'Oñon',
    cuisine: 'Mediterránea',
    name: 'Severo Manjar',
    restaurant_id: '12345678',
    grades: [
      { date: ISODate('2026-02-26T08:27:15.339Z'), grade: 'A', score: 15 },
      {
        date: ISODate('2026-02-26T08:33:21.945Z'),
        grade: 'A',
        score: 10000000
      }
    ]
  }
]
```

12) Añade un nuevo campo horario con el valor que consideres.

```
db.restaurants.updateOne({restaurant_id: '12345678'}, {$set:{"inahuguration"
: new Date("2025-2-25")}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants>  db.restaurants.updateOne({restaurant_id: '12345678'}, {$set:{"inahuguration" : new Date("2025-2-25")}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants>  db.restaurants.find({restaurant_id: '12345678'})
[
  {
    _id: ObjectId('699ffafcebf2fd6e4a628ca3'),
    address: { building: '2', coord: [ 34, 5 ], street: 'Oñon', zipcode: '33610' },
    borough: 'Oñon',
    cuisine: 'Mediterránea',
    name: 'Severo Manjar',
    restaurant_id: '12345678',
    grades: [
      { date: ISODate('2026-02-26T08:27:15.339Z'), grade: 'A', score: 15 },
      {
        date: ISODate('2026-02-26T08:33:21.945Z'),
        grade: 'A',
        score: 10000000
      }
    ],
    inahuguration: ISODate('2025-02-24T23:00:00.000Z')
  }
]
```

13) Incrementa todas las calificaciones en un punto.

```
b.restaurants.updateOne({restaurant_id: '12345678'},
{$inc:{"grades.$[].score" : 1}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants>  db.restaurants.updateOne({restaurant_id: '12345678'}, {$inc:{"grades.$[].score" : 1}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants>  db.restaurants.find({restaurant_id: '12345678'})
[
  {
    _id: ObjectId('699ffafcebf2fd6e4a628ca3'),
    address: { building: '2', coord: [ 34, 5 ], street: 'Oñon', zipcode: '33610' },
    borough: 'Oñon',
    cuisine: 'Mediterránea',
    name: 'Severo Manjar',
    restaurant_id: '12345678',
    grades: [
      { date: ISODate('2026-02-26T08:27:15.339Z'), grade: 'A', score: 16 },
      {
        date: ISODate('2026-02-26T08:33:21.945Z'),
        grade: 'A',
        score: 10000001
      }
    ],
    inahuguration: ISODate('2025-02-24T23:00:00.000Z')
  }
]
```

3) Apartado C

Escribe los comandos necesarios para realizar las siguientes operaciones de arrays sobre la colección zips:

1) Modifica los documentos de España y añade un atributo tags que contenga un array vacío.

```
db.zips.updateMany({state: 'España'}, {$set: {tags: []}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.updateMany({state: 'España'}, {$set: {tags: []}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({state: 'España'})
[
  {
    _id: ObjectId('699d5949dd4a3d6c22628ca0'),
    city: 'OVIEDO',
    zip: '33001',
    loc: { x: 43.3625025223589, y: -5.843830860844533 },
    pop: 228870,
    state: 'España',
    provincia: 'Asturias',
    tags: []
  }
]
```

2) Añade a todos los documentos que tenga el atributo tags el valor life dentro del array.

```
db.zips.updateMany({tags: {$exists: true}}, {$push: {tags: "life"}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.updateMany({tags: {$exists: true}}, {$push: {tags: "life"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({state: 'España'})
[
  {
    _id: ObjectId('699d5949dd4a3d6c22628ca0'),
    city: 'OVIEDO',
    zip: '33001',
    loc: { x: 43.3625025223589, y: -5.843830860844533 },
    pop: 228870,
    state: 'España',
    provincia: 'Asturias',
    tags: [ 'life' ]
  }
]
```

3) Modifica todos los documentos de la provincia de Alicante y añade al atributo tags el valor sun.

```
db.zips.updateMany({provincia: 'Asturias'}, {$push: {tags: "sun"}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.updateMany({provincia: 'Asturias'}, {$push: {tags: "sun"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({state: 'España'})
[
  {
    _id: ObjectId('699d5949dd4a3d6c22628ca0'),
    city: 'OVIEDO',
    zip: '33001',
    loc: { x: 43.3625025223589, y: -5.843830860844533 },
    pop: 228870,
    state: 'España',
    provincia: 'Asturias',
    tags: [ 'life', 'sun' ]
  }
]
```

4) Añade al final de tags el atributo food a todas las ciudades de España.

```
db.zips.updateMany({state: 'España', city: {$exists: true}}, {$push: {tags:
"food"}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.updateMany({state: 'España', city: {$exists: true}}, {$push: {tags: "food"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({state: 'España'})
[
  {
    _id: ObjectId('699d5949dd4a3d6c22628ca0'),
    city: 'OVIEDO',
    zip: '33001',
    loc: { x: 43.3625025223589, y: -5.843830860844533 },
    pop: 228870,
    state: 'España',
    provincia: 'Asturias',
    tags: [ 'life', 'sun', 'food' ]
  }
]
```

5) Mediante una operación, añade un nuevo atributo tourism con los valores sun y food a todos los documentos de España.

```
db.zips.updateMany({state: 'España'}, {$set: {tourism: ["sun", "food"]}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.updateMany({state: 'España'}, {$set: {tourism: ["sun", "food"]}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({state: 'España'})
[
  {
    _id: ObjectId('699d5949dd4a3d6c22628ca0'),
    city: 'OVIEDO',
    zip: '33001',
    loc: { x: 43.3625025223589, y: -5.843830860844533 },
    pop: 228870,
    state: 'España',
    provincia: 'Asturias',
    tags: [ 'life', 'sun', 'food' ],
    tourism: [ 'sun', 'food' ]
  }
]
```

6) Añade en la segunda posición de todos los documentos de España el valor drfleming en la propiedad tags.

```
db.zips.updateMany({state: 'España'}, {$push: {tags: {$each: ["drfleming"],
$position: 2}}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.updateMany({state: 'España'}, {$push: {tags: {$each: ["drfleming"], $position: 2}}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({state: 'España'})
[
  {
    _id: ObjectId('699d5949dd4a3d6c22628ca0'),
    city: 'OVIEDO',
    zip: '33001',
    loc: { x: 43.3625025223589, y: -5.843830860844533 },
    pop: 228870,
    state: 'España',
    provincia: 'Asturias',
    tags: [ 'life', 'sun', 'drfleming', 'food' ],
    tourism: [ 'sun', 'food' ]
  }
]
```

7) Modifica el valor de sun de tags de tu código postal y sustitúyelo por house.

```
db.zips.updateMany({zip: '33001', tags: "sun"}, {$set: { "tags.$" : 'house'
}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.updateMany({zip: '33001', tags: "sun"}, {$set: { "tags.$" : 'house' }})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({state: 'España'})
[
  {
    _id: ObjectId('699d5949dd4a3d6c22628ca0'),
    city: 'OVIEDO',
    zip: '33001',
    loc: { x: 43.3625025223589, y: -5.843830860844533 },
    pop: 228870,
    state: 'España',
    provincia: 'Asturias',
    tags: [ 'life', 'house', 'drfleming', 'food' ],
    tourism: [ 'sun', 'food' ]
  }
]
```

8) Elimina el valor house de tags de tu código postal.

```
db.zips.updateMany({zip: '33001', tags: "house"}, {$pull: {tags: "house"}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.updateMany({zip: '33001', tags: "house"}, {$pull: {tags: "house"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({state: 'España'})
[
  {
    _id: ObjectId('699d5949dd4a3d6c22628ca0'),
    city: 'OVIEDO',
    zip: '33001',
    loc: { x: 43.3625025223589, y: -5.843830860844533 },
    pop: 228870,
    state: 'España',
    provincia: 'Asturias',
    tags: [ 'life', 'drfleming', 'food' ],
    tourism: [ 'sun', 'food' ]
  }
]
```

9) Elimina el último valor de tags.

```
db.zips.updateMany({zip: '33001'}, {$pop: {tags: 1}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.updateMany({zip: '33001'}, {$pop: {tags: 1}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({state: 'España'})
[
  {
    _id: ObjectId('699d5949dd4a3d6c22628ca0'),
    city: 'OVIEDO',
    zip: '33001',
    loc: { x: 43.3625025223589, y: -5.843830860844533 },
    pop: 228870,
    state: 'España',
    provincia: 'Asturias',
    tags: [ 'life', 'drfleming' ],
    tourism: [ 'sun', 'food' ]
  }
]
```

10) Elimina el primer valor de tourism.

```
db.zips.updateMany({zip: '33001'}, {$pop: {tourism: -1}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.updateMany({zip: '33001'}, {$pop: {tourism: -1}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-hymc7m-shard-0 [primary] sample_training> db.zips.find({state: 'España'})
[
  {
    _id: ObjectId('699d5949dd4a3d6c22628ca0'),
    city: 'OVIEDO',
    zip: '33001',
    loc: { x: 43.3625025223589, y: -5.843830860844533 },
    pop: 228870,
    state: 'España',
    provincia: 'Asturias',
    tags: [ 'life', 'drfleming' ],
    tourism: [ 'food' ]
  }
]
```

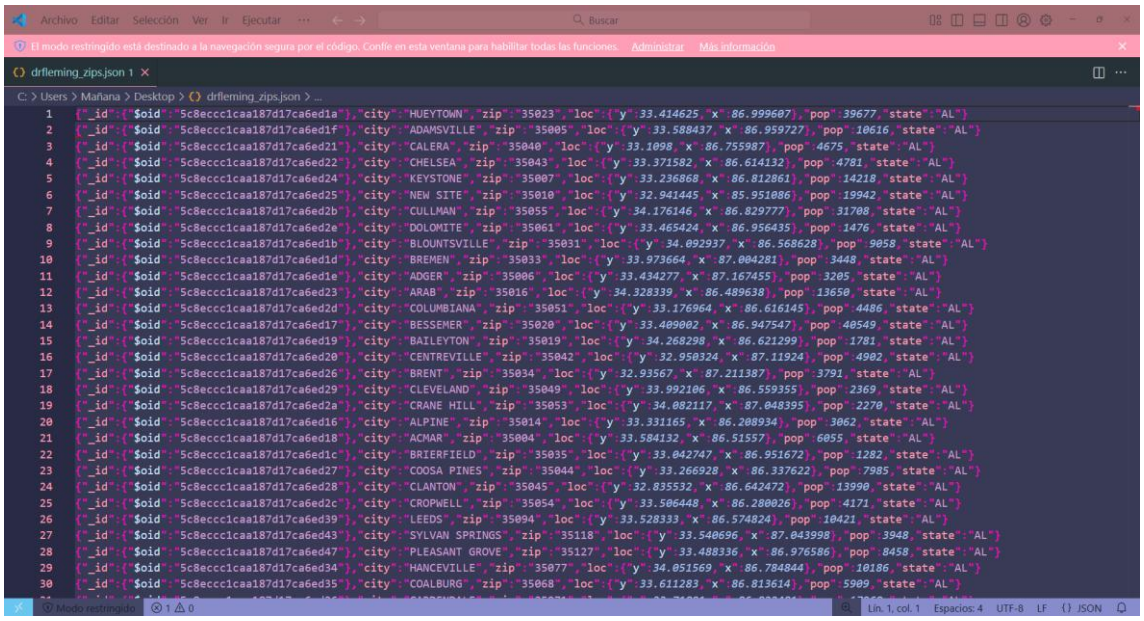
4) Apartado D

Utilizando las herramientas de administración de MongoDB (las cuales puedes utilizarlas desde dentro de Docker o bien descargar las MongoTools), se pide:

1) Exporta los datos de la colección zips en drfleming_zips.json

```
mongoexport /uri:"mongodb+srv://cluster0.0251v4q.mongodb.net/"
/username:pablo /password:auuuuu123 /db:sample_training /collection:zips
/out:"C:\Users\Mañana\Desktop\drfleming_zips.json"
```

```
2026-03-02T10:18:11.177+0100    [.......................]  sample_training.zips  0/29471  (0.0%)
2026-03-02T10:18:12.191+0100    [.......................]  sample_training.zips  0/29471  (0.0%)
2026-03-02T10:18:13.177+0100    [.......................]  sample_training.zips  0/29471  (0.0%)
2026-03-02T10:18:14.177+0100    [.......................]  sample_training.zips  0/29471  (0.0%)
2026-03-02T10:18:15.176+0100    [.......................]  sample_training.zips  0/29471  (0.0%)
2026-03-02T10:18:16.183+0100    [.......................]  sample_training.zips  0/29471  (0.0%)
2026-03-02T10:18:17.177+0100    [.......................]  sample_training.zips  0/29471  (0.0%)
2026-03-02T10:18:18.191+0100    [.......................]  sample_training.zips  0/29471  (0.0%)
2026-03-02T10:18:19.110+0100    [#######################]  sample_training.zips  29471/29471  (100.0%)
2026-03-02T10:18:19.110+0100    exported 29471 records
```

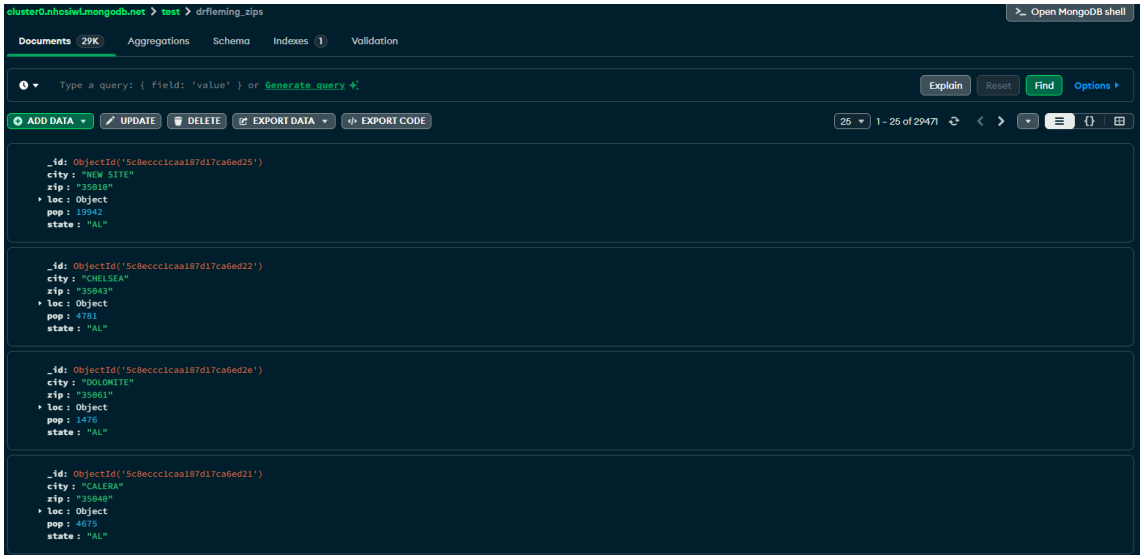


2) Importa la colección recién exportada dentro de la base de datos pruebas en la colección drfleming_zips.

```
mongoimport.exe /uri:"mongodb+srv://cluster0.0251v4q.mongodb.net/"
/username:pablo /password:auuuuu123 /db:test /collection:drfleming_zips
/file:"C:\Users\Mañana\Desktop\drfleming_zips.json"
```

```
2026-03-03T08:41:08.963+0100    [#######################..] test.drfleming_zips  3.65MB/3.83MB (95.2%)
2026-03-03T08:41:11.963+0100    [########################.] test.drfleming_zips  3.78MB/3.83MB (98.6%)
2026-03-03T08:41:14.557+0100    [########################] test.drfleming_zips  3.83MB/3.83MB (100.0%)
2026-03-03T08:41:14.557+0100    29471 document(s) imported successfully. 0 document(s) failed to import.
```

3) Realiza una consulta y comprueba que los datos se han importado correctamente.



4) Exporta la base de datos de sample_trainings y muestra el resultado.

```
mongodump.exe /uri:"mongodb+srv://cluster0.0251v4q.mongodb.net/"
/username:pablo /password:auuuuu123 /db:drfleming
/out:"C:\Users\Mañana\Desktop\Mondongos\drfleming_db"
```

Lo haré con drfleming para no tardar cuatro horas.

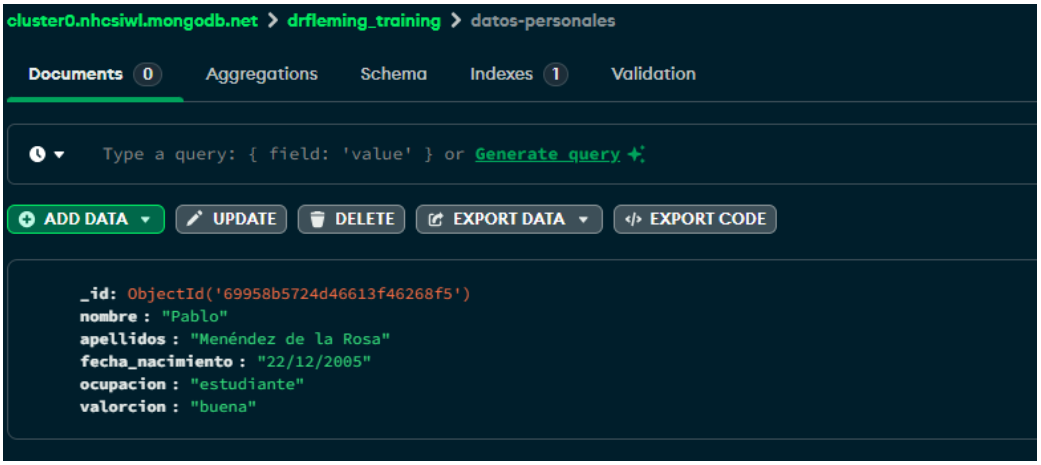
```
2026-03-03T08:52:17.468+0100    writing drfleming.datos-personales to C:\Users\Mañana\Desktop\Mondongos\drfleming_db\drfleming\datos-personales.bson
2026-03-03T08:52:17.897+0100    done dumping drfleming.datos-personales (1 document)
```

5) Importa la base de datos recién exportada en drfleming_trainings

```
mongorestore.exe /uri:"mongodb+srv://cluster0.0251v4q.mongodb.net/"
/username:pablo /password:auuuuu123 --db=drfleming_training
/dir:"C:\Users\Mañana\Desktop\Mondongos\drfleming"
```

```
2026-03-03T09:14:09.610+0100    The --db and --collection flags are deprecated for this use-case; please use --nsInclude instead, i.e. with --nsInclude=${DATABASE}.${COLLE
CTION}
2026-03-03T09:14:09.633+0100    building a list of collections to restore from C:\Users\Mañana\Desktop\Mondongos\drfleming dir
2026-03-03T09:14:09.635+0100    don't know what to do with file "C:\Users\Mañana\Desktop\Mondongos\drfleming\prelude.json", skipping...
2026-03-03T09:14:09.635+0100    reading metadata for drfleming_training.datos-personales from C:\Users\Mañana\Desktop\Mondongos\drfleming\datos-personales.metadata.json
2026-03-03T09:14:09.700+0100    restoring to existing collection drfleming_training.datos-personales without dropping
2026-03-03T09:14:09.700+0100    restoring drfleming_training.datos-personales from C:\Users\Mañana\Desktop\Mondongos\drfleming\datos-personales.bson
2026-03-03T09:14:10.058+0100    finished restoring drfleming_training.datos-personales (1 document, 0 failures)
2026-03-03T09:14:10.059+0100    no indexes to restore for collection drfleming_training.datos-personales
2026-03-03T09:14:10.060+0100    1 document(s) restored successfully. 0 document(s) failed to restore.
```

6) Realiza una consulta y comprueba que los datos se han importado correctamente.

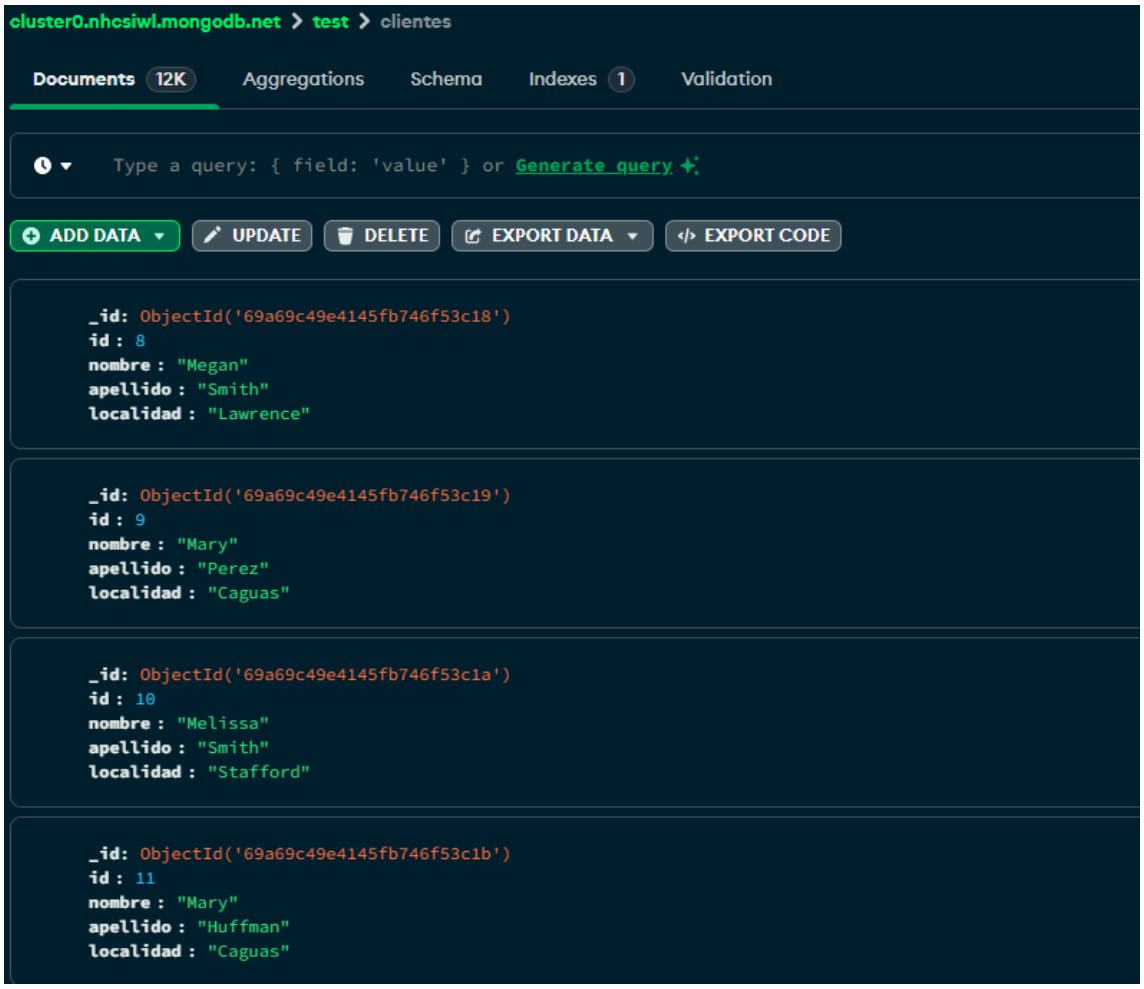


7) Importa el archivo clientes.csv dentro de la base de datos pruebas en la colección clientes.

```
mongoimport.exe /uri:"mongodb+srv://cluster0.0251v4q.mongodb.net/"
/username:pablo /password:auuuuu123 /db:test /collection:clientes
/file:"C:\Users\Mañana\Desktop\Datasets\clientes.csv"
/fields:id,nombre,apellido,localidad /type:csv
```

```
2026-03-03T09:29:16.301+0100    [###############.........] test.clientes        216KB/337KB (64.1%)
2026-03-03T09:29:19.301+0100    [##################.......] test.clientes        244KB/337KB (72.5%)
2026-03-03T09:29:22.300+0100    [####################.....] test.clientes        272KB/337KB (80.8%)
2026-03-03T09:29:25.300+0100    [######################...] test.clientes        300KB/337KB (89.1%)
2026-03-03T09:29:28.299+0100    [#######################.] test.clientes        328KB/337KB (97.4%)
2026-03-03T09:29:31.299+0100    [########################] test.clientes        337KB/337KB (100.0%)
2026-03-03T09:29:32.206+0100    [########################] test.clientes        337KB/337KB (100.0%)
```

8) Realiza una consulta y comprueba que los datos se han importado correctamente.



9) Importa el archivo poblacion.tsv dentro de la base de datos pruebas en la colección poblacion.

```
mongoimport.exe /uri:"mongodb+srv://cluster0.0251v4q.mongodb.net/"
/username:pablo /password:auuuuu123 /db:test /collection:poblacion
/headerline /file:"C:\Users\Mañana\Desktop\Datasets\poblacion.tsv" /type:tsv
```

```
2026-03-03T09:35:18.894+0100    [####################....] test.poblacion       296KB/342KB (86.4%)
2026-03-03T09:35:21.893+0100    [####################....] test.poblacion       296KB/342KB (86.4%)
2026-03-03T09:35:24.893+0100    [#######################.] test.poblacion       340KB/342KB (99.3%)
2026-03-03T09:35:25.777+0100    [########################] test.poblacion       342KB/342KB (100.0%)
2026-03-03T09:35:25.777+0100    8116 document(s) imported successfully. 0 document(s) failed to import.
```

10) Realiza una consulta y comprueba que los datos se han importado correctamente.

cluster0.nhcsiwl.mongodb.net > test > poblacion

Documents 8.1K | Aggregations | Schema | Indexes 1

Type a query: { field: 'value' } or Generate query

ADD DATA | UPDATE | DELETE | EXPORT DATA

```
_id: ObjectId('69a69d350cc7ccb42221c2ec')
CPRO : 2
PROVINCIA : "Albacete"
CMUN : 2
NOMBRE : "Alatoz"
POB13 : 572
HOMBRES : 318
MUJERES : 254

_id: ObjectId('69a69d350cc7ccb42221c2ed')
CPRO : 2
PROVINCIA : "Albacete"
CMUN : 1
NOMBRE : "Abengibre"
POB13 : 819
HOMBRES : 409
MUJERES : 410

_id: ObjectId('69a69d350cc7ccb42221c2ee')
CPRO : 2
PROVINCIA : "Albacete"
CMUN : 4
NOMBRE : "Albatana"
POB13 : 774
HOMBRES : 394
MUJERES : 380

_id: ObjectId('69a69d350cc7ccb42221c2ef')
CPRO : 2
PROVINCIA : "Albacete"
CMUN : 3
NOMBRE : "Albacete"
POB13 : 172.693
HOMBRES : 84.702
MUJERES : 87.991
```

5) Apartado E

Escribe los comandos necesarios para realizar las siguientes consultas agregadas sobre la colección sample_supplies.sales:

1) Cantidad total de ventas registradas.

```
db.sales.find().count()
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_supplies> db.sales.find().count()
5000
```

2) Cantidad total de ventas por método de compra (purchaseMethod).

```
db.sales.distinct('purchaseMethod').length
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_supplies> db.sales.distinct('purchaseMethod').length
3
```

3) Investiga el operador $unwind y calcula el ingreso total de cada producto (name dentro de items)

```
db.sales.aggregate([{$unwind: "$items"}, {$group: {_id: "$items.name",
totalIncome: { $sum: {$multiply: ["$items.price", "$items.quantity"]}}}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_supplies> db.sales.aggregate([{$unwind: "$items"}, {$group: {_id: "$items.name", totalIncome: { $sum: {$multiply: ["$items.pric
e", "$items.quantity"]}}}}])
[
  { _id: 'notepad', totalIncome: Decimal128('463615.48') },
  { _id: 'binder', totalIncome: Decimal128('511644.57') },
  { _id: 'laptop', totalIncome: Decimal128('6775977.07') },
  { _id: 'pens', totalIncome: Decimal128('581843.27') },
  { _id: 'envelopes', totalIncome: Decimal128('376658.49') },
  { _id: 'printer paper', totalIncome: Decimal128('367459.29') },
  { _id: 'backpack', totalIncome: Decimal128('817374.18') }
]
```

4) Calcular el promedio de elementos por venta.

```
db.sales.aggregate([{$project: {numItems: {$size: "$items"}}},{$group: {_id:
null,avgItemsPerSale: {$avg: "$numItems"}}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_supplies> db.sales.aggregate([{$project: {numItems: {$size: "$items"}}},{$group: {_id: null,avgItemsPerSale: {$avg: "$numItems"
}}}])
[ { _id: null, avgItemsPerSale: 5.4876 } ]
```

5) A partir de todos los elementos, calcula el ingreso total por ventas, teniendo en cuenta el precio y la cantidad de cada elemento.

```
db.sales.aggregate([{$unwind:'$items'},{$project:{ingreso:{$multiply:['$items
.price','$items.quantity']}}},{$group:{_id:null,ingresoTotal:{$sum:'$ingreso'
}}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_supplies> db.sales.aggregate([{$unwind:'$items'},{$project:{ingreso:{$multiply:['$items.price','$items.quantity']}}},{$group:{_
id:null,ingresoTotal:{$sum:'$ingreso'}}}])
[ { _id: null, ingresoTotal: Decimal128('9894572.27') } ]
```

6) Encontrar los 3 productos más vendidos (por cantidad de unidades vendidas).

```
db.sales.aggregate([{$unwind:'$items'},{$group:{_id:'$items.name',totalSold:{
$sum:'$items.quantity'}}},{$sort:{totalSold:-1}},{$limit:3}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_supplies> db.sales.aggregate([{$unwind:'$items'},{$group:{_id:'$items.name',totalSold:{$sum:'$items.quantity'}}},{$sort:{totalS
old:-1}},{$limit:3}])
[
  { _id: 'binder', totalSold: 25493 },
  { _id: 'envelopes', totalSold: 25078 },
  { _id: 'notepad', totalSold: 20727 }
]
```

7) Averiguar cuantas ventas se han realizado del producto notepad.

```
db.sales.aggregate([{$unwind:'$items'},{$match:{'items.name':'notepad'}},{$co
unt:'ventas_notepad'}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_supplies> db.sales.aggregate([{$unwind:'$items'},{$match:{'items.name':'notepad'}},{$count:'ventas_notepad'}])
[ { ventas_notepad: 6872 } ]
```

8) Averiguar cuantos notepad se han vendido y cuanto se ha recaudado por sus ventas.

```
db.sales.aggregate([{$unwind:'$items'},{$match:{'items.name':'notepad'}},{$gr
oup:{_id:null,totalNotepadsSold:{$sum:'$items.quantity'},totalRevenue:{$sum:{
$multiply:['$items.price','$items.quantity']}}}},{$project:{_id:0,totalNotepa
dsSold:1,totalRevenue:1}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_supplies> db.sales.aggregate([{$unwind:'$items'},{$match:{'items.name':'notepad'}},{$group:{_id:null,totalNotepadsSold:{$sum:'$
items.quantity'},totalRevenue:{$sum:{$multiply:['$items.price','$items.quantity']}}}},{$project:{_id:0,totalNotepadsSold:1,totalRevenue:1}}])
[ { totalNotepadsSold: 20727, totalRevenue: Decimal128('463615.48') } ]
```

9) Averiguar los tres productos con un precio medio más alto.

```
db.sales.aggregate([{$unwind:'$items'},{$group:{_id:'$items.name',avgPrice:{$
avg:'$items.price'}}},{$sort:{avgPrice:-1}},{$limit:3}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_supplies> db.sales.aggregate([{$unwind:'$items'},{$group:{_id:'$items.name',avgPrice:{$avg:'$items.price'}}},{$sort:{avgPrice:-
1}},{$limit:3}])
[
  {
    _id: 'laptop',
    avgPrice: Decimal128('992.15200792602377807133421400026420')
  },
  {
    _id: 'backpack',
    avgPrice: Decimal128('118.40834127327847553053269813772 20')
  },
  {
    _id: 'pens',
    avgPrice: Decimal128('42.23900636104408861592454485632814')
  }
]
```

10) Averiguar la cantidad de hombres y mujeres que han realizado compras, así como sus edades medias.

```
db.sales.aggregate([{$group:{_id:'$customer.gender',cantidad:{$sum:1},edad_me
dia:{$avg:'$customer.age'}}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_supplies> db.sales.aggregate([{$group:{_id:'$customer.gender',cantidad:{$sum:1},edad_media:{$avg:'$customer.age'}}}])
[
  { _id: 'F', cantidad: 2527, edad_media: 45.015433320014246 },
  { _id: 'M', cantidad: 2473, edad_media: 44.449656287909 42 }
]
```

11) Recuperar el nombre y el email de los 5 clientes que más han gastado.

```
db.sales.aggregate([{$unwind:'$items'},{$group:{_id:{name:'$customer.email',e
mail:'$customer.email'},totalSpent:{$sum:{$multiply:['$items.price','$items.q
uantity']}},name:{$first:'$customer.email'},email:{$first:'$customer.email'}}
},{$sort:{totalSpent:-
1}},{$limit:5},{$project:{_id:0,name:'$name',email:'$email'}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_supplies> db.sales.aggregate([{$unwind:'$items'},{$group:{_id:{name:'$customer.email',email:'$customer.email'},totalSpent:{$sum
:{$multiply:['$items.price','$items.quantity']}},name:{$first:'$customer.email'},email:{$first:'$customer.email'}}},{$sort:{totalSpent:-1}},{$limit:5},{$project:{_id:0,nam
e:'$name',email:'$email'}}])
[
  { name: 'kegzuh@bok.hk', email: 'kegzuh@bok.hk' },
  { name: 'mebni@juubku.mo', email: 'mebni@juubku.mo' },
  { name: 'upi@iwtubfa.km', email: 'upi@iwtubfa.km' },
  { name: 'woud@argav.hm', email: 'woud@argav.hm' },
  { name: 'li@nico.lb', email: 'li@nico.lb' }
]
```

12) Para cada email, mostrar la cantidad de ventas realizadas, así como la cantidad de productos adquiridos.

```
db.sales.aggregate([{$unwind:'$items'},{$group:{_id:'$customer.email',totalSales:{$addToSet:'$_id'},totalProducts:{$sum:'$items.quantity'}}},{$project:{email:'$_id',totalSales:{$size:'$totalSales'},totalProducts:1,_id:0}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_supplies> db.sales.aggregate([{$unwind:'$items'},{$group:{_id:'$customer.email',totalSales:{$addToSet:'$_id'},totalProducts:{$sum:'$items.quantity'}}},{$project:{email:'$_id',totalSales:{$size:'$totalSales'},totalProducts:1,_id:0}}])
[
  { totalProducts: 35, email: 'paj@mouvzi.fi', totalSales: 1 },
  { totalProducts: 19, email: 'dukum@atasuimo.mz', totalSales: 1 },
  { totalProducts: 31, email: 'zasiej@za.mz', totalSales: 1 },
  { totalProducts: 12, email: 'jiseanu@vozre.ac', totalSales: 1 },
  { totalProducts: 15, email: 'gi@meli.np', totalSales: 1 },
  { totalProducts: 40, email: 'puc@dev.nf', totalSales: 1 },
  { totalProducts: 8, email: 'ge@ji.nr', totalSales: 1 },
  { totalProducts: 24, email: 'nimu@fokvodfa.sm', totalSales: 1 },
  { totalProducts: 12, email: 'sa@ve.my', totalSales: 1 },
  { totalProducts: 33, email: 'seaduci@cevohhip.ug', totalSales: 1 },
  { totalProducts: 34, email: 'baftolmav@wocelov.gr', totalSales: 1 },
  { totalProducts: 29, email: 'sunsa@zanjibar.je', totalSales: 1 },
  { totalProducts: 25, email: 'vozvicna@ge.co', totalSales: 1 },
  { totalProducts: 22, email: 'vom@gacair.to', totalSales: 1 },
  { totalProducts: 18, email: 'adi@zusnuzgu.mo', totalSales: 1 },
  { totalProducts: 31, email: 'pu@pagmuga.bv', totalSales: 1 },
  { totalProducts: 3, email: 'jutteuna@gaka.cz', totalSales: 1 },
  { totalProducts: 8, email: 'go@valda.sd', totalSales: 1 },
  { totalProducts: 15, email: 'cem@ginjehu.va', totalSales: 1 },
  { totalProducts: 5, email: 'pit@ujelinhe.bt', totalSales: 1 }
]
```

13) Recupera cual es el producto que más se ha vendido en London.

```
db.sales.aggregate([{$match:{storeLocation:'London'}},{$unwind:'$items'},{$group:{_id:'$items.name',totalSold:{$sum:'$items.quantity'}}},{$sort:{totalSold:-1}},{$limit:1}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_supplies> db.sales.aggregate([{$match:{storeLocation:'London'}},{$unwind:'$items'},{$group:{_id:'$items.name',totalSold:{$sum:'$items.quantity'}}},{$sort:{totalSold:-1}},{$limit:1}])
[ { _id: 'envelopes', totalSold: 4037 } ]
```

14) Obtener el nombre y el email del cliente, y el total gastado, pero solo para aquellos clientes que han gastado más de 1000$.

```
db.sales.aggregate([{$unwind:'$items'},{$group:{_id:{name:'$customer.email',email:'$customer.email'},totalSpent:{$sum:{$multiply:['$items.price','$items.quantity']}},name:{$first:'$customer.email'},email:{$first:'$customer.email'}}},{$match:{totalSpent:{$gt:1000}}},{$project:{_id:0,name:'$name',email:'$email',totalSpent:1}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_supplies> db.sales.aggregate([{$unwind:'$items'},{$group:{_id:{name:'$customer.email',email:'$customer.email'},totalSpent:{$sum:{$multiply:['$items.price','$items.quantity']}},name:{$first:'$customer.email'},email:{$first:'$customer.email'}}},{$match:{totalSpent:{$gt:1000}}},{$project:{_id:0,name:'$name',email:'$email',totalSpent:1}}])
[
  {
    totalSpent: Decimal128('6703.97'),
    name: 'popsek@ahu.ky',
    email: 'popsek@ahu.ky'
  },
  {
    totalSpent: Decimal128('2221.36'),
    name: 'vud@gobkidwa.gi',
    email: 'vud@gobkidwa.gi'
  },
  {
    totalSpent: Decimal128('1175.04'),
    name: 'dusivu@he.om',
    email: 'dusivu@he.om'
  },
  {
    totalSpent: Decimal128('1105.10'),
    name: 'ijugel@baj.pw',
    email: 'ijugel@baj.pw'
  },
  {
```

15) Averiguar cuáles han sido los tres años que más ventas se han realizado.

```
db.sales.aggregate([{$group:{_id:{$year:'$saleDate'},totalSales:{$sum:1}}},{$sort:{totalSales:-1}},{$limit:3}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_supplies> db.sales.aggregate([{$group:{_id:{$year:'$saleDate'},totalSales:{$sum:1}}},{$sort:{totalSales:-1}},{$limit:3}])
[
  { _id: 2017, totalSales: 1052 },
  { _id: 2016, totalSales: 1010 },
  { _id: 2015, totalSales: 1006 }
]
```

6) Apartado F

Escribe los comandos necesarios para realizar las siguientes consultas agregadas sobre la colección sample_restaurants.restaurants:

1) Cantidad total de restaurantes.

```
db.restaurants.aggregate([{$count: "total_restaurants"}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants> db.restaurants.aggregate([{$count: "total_restaurants"}])
[ { total_restaurants: 25360 } ]
```

2) Cantidad de restaurantes por barrio (propiedad borough).

```
db.restaurants.aggregate([{$group:{_id:'$borough',count:{$sum:1}}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants> db.restaurants.aggregate([{$group:{_id:'$borough',count:{$sum:1}}}])
[
  { _id: 'Brooklyn', count: 6086 },
  { _id: 'Manhattan', count: 10259 },
  { _id: 'Staten Island', count: 969 },
  { _id: 'Bronx', count: 2338 },
  { _id: 'Oñon', count: 1 },
  { _id: 'Queens', count: 5656 },
  { _id: 'Missing', count: 51 }
]
```

3) Recupera los cinco barrios con mayor cantidad de restaurantes.

```
db.restaurants.aggregate([{$group:{_id:'$borough',count:{$sum:1}}},{$sort:{co
unt:-1}},{$limit:5}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants> db.restaurants.aggregate([{$group:{_id:'$borough',count:{$sum:1}}},{$sort:{count:-1}},{$limit:5}])
[
  { _id: 'Manhattan', count: 10259 },
  { _id: 'Brooklyn', count: 6086 },
  { _id: 'Queens', count: 5656 },
  { _id: 'Bronx', count: 2338 },
  { _id: 'Staten Island', count: 969 }
]
```

4) Investiga el operador $unwind, y recupera la calificación media del restaurante Palm Restauran.

```
db.restaurants.aggregate([{$match:{name:'Palm
Restauran'}},{$unwind:'$grades'},{$group:{_id:'$name',averageScore:{$avg:'$gr
ades.score'}}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants> db.restaurants.aggregate([{$match:{name:'Palm Restauran'}},{$unwind:'$grades'},{$group:{_id:'$name',averageScore:{
$avg:'$grades.score'}}}])
```

5) Recupera la fecha de la última calificación de cada restaurante.

```
db.restaurants.aggregate([{$unwind:'$grades'},{$sort:{restaurant_id:1,'grades
.date':-
1}},{$group:{_id:'$restaurant_id',name:{$first:'$name'},lastGradeDate:{$first
:'$grades.date'}}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants> db.restaurants.aggregate([{$unwind:'$grades'},{$sort:{restaurant_id:1,'grades.date':-1}},{$group:{_id:'$restaurant
_id',name:{$first:'$name'},lastGradeDate:{$first:'$grades.date'}}}])
MongoServerError[QueryExceededMemoryLimitNoDiskUseAllowed]: PlanExecutor error during aggregation :: caused by :: Sort exceeded memory limit of 33554432 bytes, but did not
 opt in to external sorting.
```

6) Encuentra todos los restaurantes que al menos tienen una calificación de A.

```
db.restaurants.aggregate([{$match:{'grades.grade':'A'}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants> db.restaurants.aggregate([{$match:{'grades.grade':'A'}}])
[
  {
    _id: ObjectId('5eb3d668b31de5d588f4292b'),
    address: {
      building: '7114',
      coord: [ -73.9068506, 40.6199034 ],
      street: 'Avenue U',
      zipcode: '11234'
    },
    borough: 'Brooklyn',
    cuisine: 'Delicatessen',
    grades: [
      { date: ISODate('2014-05-29T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2014-01-14T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2013-08-03T00:00:00.000Z'), grade: 'A', score: 8 },
      { date: ISODate('2012-07-18T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2012-03-09T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2011-10-14T00:00:00.000Z'), grade: 'A', score: 9 }
    ],
    name: "Wilken'S Fine Food",
    restaurant_id: '40356483'
  },
  {
    _id: ObjectId('5eb3d668b31de5d588f4292f'),
    address: {
      building: '97-22',
      coord: [ -73.8601152, 40.7311739 ],
      street: '63 Road',
      zipcode: '11374'
    },
    borough: 'Queens',
    cuisine: 'Jewish/Kosher',
    grades: [
      { date: ISODate('2014-11-24T00:00:00.000Z'), grade: 'Z', score: 20 },
      { date: ISODate('2013-01-17T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2012-08-02T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2011-12-15T00:00:00.000Z'), grade: 'B', score: 25 }
```

7) Mediante $unwind, cuenta cuántas calificaciones de tipo A, B, C, etc.... existen en total en todos los restaurantes.

```
db.restaurants.aggregate([{$unwind:'$grades'},{$group:{_id:'$grades.grade',count:{$sum:1}}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants> db.restaurants.aggregate([{$unwind:'$grades'},{$group:{_id:'$grades.grade',count:{$sum:1}}}])
[
  { _id: 'A', count: 74658 },
  { _id: 'Z', count: 1337 },
  { _id: 'B', count: 12603 },
  { _id: 'C', count: 3145 },
  { _id: 'P', count: 1197 },
  { _id: 'Not Yet Graded', count: 525 }
]
```

8) Mostrar los restaurantes que tienen más de tres calificaciones.

```
db.restaurants.aggregate([{$addFields:{gradesCount:{$size:'$grades'}}},{$match:{gradesCount:{$gt:3}}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants> db.restaurants.aggregate([{$addFields:{gradesCount:{$size:'$grades'}}},{$match:{gradesCount:{$gt:3}}}])
[
  {
    _id: ObjectId('5eb3d668b31de5d588f4292b'),
    address: {
      building: '7114',
      coord: [ -73.9068506, 40.6199034 ],
      street: 'Avenue U',
      zipcode: '11234'
    },
    borough: 'Brooklyn',
    cuisine: 'Delicatessen',
    grades: [
      { date: ISODate('2014-05-29T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2014-01-14T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2013-08-03T00:00:00.000Z'), grade: 'A', score: 8 },
      { date: ISODate('2012-07-18T00:00:00.000Z'), grade: 'A', score: 10 },
      { date: ISODate('2012-03-09T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2011-10-14T00:00:00.000Z'), grade: 'A', score: 9 }
    ],
    name: "Wilken'S Fine Food",
    restaurant_id: '40356483',
    gradesCount: 6
  },
  {
    _id: ObjectId('5eb3d668b31de5d588f4292f'),
    address: {
      building: '97-22',
      coord: [ -73.8601152, 40.7311739 ],
      street: '63 Road',
      zipcode: '11374'
    },
    borough: 'Queens',
    cuisine: 'Jewish/Kosher',
    grades: [
      { date: ISODate('2014-11-24T00:00:00.000Z'), grade: 'Z', score: 20 },
      { date: ISODate('2013-01-17T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2012-08-02T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2011-12-15T00:00:00.000Z'), grade: 'B', score: 25 }
    ],
    name: 'Tov Kosher Kitchen',
    restaurant_id: '40356068',
    gradesCount: 4
```

9) Mostrar los restaurantes que tienen más de cuatro calificaciones y que hagan cocina American.

```
db.restaurants.aggregate([{$match:{cuisine:'American'}},{$addFields:{gradesCount:{$size:'$grades'}}},{$match:{gradesCount:{$gt:4}}},{$project:{gradesCount:0}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants> db.restaurants.aggregate([{$match:{cuisine:'American'}},{$addFields:{gradesCount:{$size:'$grades'}}},{$match:{gradesCount:{$gt:4}}},{$project:{gradesCount:0}}])
[
  {
    _id: ObjectId('5eb3d668b31de5d588f42940'),
    address: {
      building: '522',
      coord: [ -73.95171, 40.767461 ],
      street: 'East   74 Street',
      zipcode: '10021'
    },
    borough: 'Manhattan',
    cuisine: 'American',
    grades: [
      { date: ISODate('2014-09-02T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2013-12-19T00:00:00.000Z'), grade: 'B', score: 16 },
      { date: ISODate('2013-05-28T00:00:00.000Z'), grade: 'A', score: 9 },
      { date: ISODate('2012-12-07T00:00:00.000Z'), grade: 'A', score: 13 },
      { date: ISODate('2012-03-29T00:00:00.000Z'), grade: 'A', score: 11 }
    ],
    name: 'Glorious Food',
    restaurant_id: '40361521'
  },
  {
    _id: ObjectId('5eb3d668b31de5d588f42931'),
    address: {
      building: '6409',
      coord: [ -74.00528899999999, 40.628886 ],
      street: '11 Avenue',
      zipcode: '11219'
    },
    borough: 'Brooklyn',
    cuisine: 'American',
    grades: [
      { date: ISODate('2014-07-18T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2013-07-30T00:00:00.000Z'), grade: 'A', score: 12 },
      { date: ISODate('2013-02-13T00:00:00.000Z'), grade: 'A', score: 11 },
      { date: ISODate('2012-08-16T00:00:00.000Z'), grade: 'A', score: 2 },
      { date: ISODate('2011-08-17T00:00:00.000Z'), grade: 'A', score: 11 }
    ],
    name: 'Regina Caterers',
    restaurant_id: '40356649'
  },
```

10) Encontrar el restaurante con la calificación A más alta (mayor score).

```
db.restaurants.aggregate([{$unwind:'$grades'},{$match:{'grades.grade':'A'}},{$sort:{'grades.score':-1}},{$limit:1}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants> db.restaurants.aggregate([{$unwind:'$grades'},{$match:{'grades.grade':'A'}},{$sort:{'grades.score':-1}},{$limit:1}
])
[
  {
    _id: ObjectId('699ffafcebf2fd6e4a628ca3'),
    address: { building: '2', coord: [ 34, 5 ], street: 'Oñon', zipcode: '33610' },
    borough: 'Oñon',
    cuisine: 'Mediterránea',
    name: 'Severo Manjar',
    restaurant_id: '12345678',
    grades: {
      date: ISODate('2026-02-26T08:33:21.945Z'),
      grade: 'A',
      score: 10000001
    },
    inahuguration: ISODate('2025-02-24T23:00:00.000Z')
  }
]
```

11) Calcular la diferencia entre el score más alto y bajo para cada restaurante que tenga al menos dos grades.

```
db.restaurants.aggregate([{$addFields:{gradesCount:{$size:'$grades'}}},{$matc
h:{gradesCount:{$gte:2}}},{$addFields:{maxScore:{$max:'$grades.score'},minSco
re:{$min:'$grades.score'}}},{$project:{name:1,restaurant_id:1,scoreDiff:{$sub
tract:['$maxScore','$minScore']}}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants> db.restaurants.aggregate([{$addFields:{gradesCount:{$size:'$grades'}}},{$match:{gradesCount:{$gte:2}}},{$addFields
:{maxScore:{$max:'$grades.score'},minScore:{$min:'$grades.score'}}},{$project:{name:1,restaurant_id:1,scoreDiff:{$subtract:['$maxScore','$minScore']}}}])
[
  {
    _id: ObjectId('5eb3d668b31de5d588f4292b'),
    name: "Wilken'S Fine Food",
    restaurant_id: '40356483',
    scoreDiff: 5
  },
  {
    _id: ObjectId('5eb3d668b31de5d588f4292f'),
    name: 'Tov Kosher Kitchen',
    restaurant_id: '40356068',
    scoreDiff: 12
  },
  {
    _id: ObjectId('5eb3d668b31de5d588f42930'),
    name: 'Brunos On The Boulevard',
    restaurant_id: '40356151',
    scoreDiff: 31
  },
  {
    _id: ObjectId('5eb3d668b31de5d588f42932'),
    name: 'Taste The Tropics Ice Cream',
    restaurant_id: '40356731',
    scoreDiff: 7
  },
```

12) Obtener el nombre del restaurante, su dirección (street y zipcode) y la última fecha de inspección (grade.date) para los restaurantes del Bronx.

```
db.restaurants.aggregate([{$match:{borough:'Bronx'}},{$addFields:{lastInspect
ion:{$max:'$grades.date'}}},{$project:{_id:0,name:1,street:'$address.street',
zipcode:'$address.zipcode',lastInspection:1}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants> db.restaurants.aggregate([{$match:{borough:'Bronx'}},{$addFields:{lastInspection:{$max:'$grades.date'}}},{$project
:{_id:0,name:1,street:'$address.street',zipcode:'$address.zipcode',lastInspection:1}}])
[
  {
    name: 'Morris Park Bake Shop',
    lastInspection: ISODate('2014-03-03T00:00:00.000Z'),
    street: 'Morris Park Ave',
    zipcode: '10462'
  },
  {
    name: 'Wild Asia',
    lastInspection: ISODate('2014-05-28T00:00:00.000Z'),
    street: 'Southern Boulevard',
    zipcode: '10460'
  },
  {
    name: 'Carvel Ice Cream',
    lastInspection: ISODate('2014-04-24T00:00:00.000Z'),
    street: 'East 233 Street',
    zipcode: '10466'
  },
  {
    name: 'Happy Garden',
    lastInspection: ISODate('2013-12-30T00:00:00.000Z'),
    street: '238 Spofford Ave',
    zipcode: '10474'
  },
  {
    name: 'Happy Garden',
    lastInspection: ISODate('2014-03-03T00:00:00.000Z'),
    street: 'East Kingsbridge Road',
    zipcode: '10458'
  },
```

13) Encontrar los 3 barrios que tienen la mayor variedad de tipos de cocina.

```
db.restaurants.aggregate([{$group:{_id:{borough:'$borough',cuisine:'$cuisine'
}}},{$group:{_id:'$_id.borough',cuisineCount:{$sum:1}}},{$sort:{cuisineCount:
-1}},{$limit:3}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] sample_restaurants> db.restaurants.aggregate([{$group:{_id:{borough:'$borough',cuisine:'$cuisine'}}},{$group:{_id:'$_id.borough',cuisi
neCount:{$sum:1}}},{$sort:{cuisineCount:-1}},{$limit:3}])
[
  { _id: 'Manhattan', cuisineCount: 82 },
  { _id: 'Brooklyn', cuisineCount: 80 },
  { _id: 'Queens', cuisineCount: 74 }
]
```

7) Apartado G

Importa el fichero mongo_cities1000.json en pruebas.cities. Sobre dichos datos, resuelve las siguientes consultas:

1) Calcula el número total de ciudades.

```
db.cities.find().count()
```

```
Atlas atlas-hymc7m-shard-0 [primary] test> db.cities.find().count()
80000
```

2) Recupera todos los datos de la ciudad cuyo nombre es Oviedo.

```
db.cities.find({name: "Oviedo"})
```

```
Atlas atlas-hymc7m-shard-0 [primary] test> db.cities.find({name: "Oviedo"})
[
  {
    _id: ObjectId('69a80345dfae595dec9d961b'),
    name: 'Oviedo',
    country: 'DO',
    timezone: 'America/Santo_Domingo',
    population: 2671,
    location: { latitude: 17.78333, longitude: -71.36667 }
  },
  {
    _id: ObjectId('69a80347dfae595dec9d9fd6'),
    name: 'Oviedo',
    country: 'ES',
    timezone: 'Europe/Madrid',
    population: 224005,
    location: { latitude: 43.36029, longitude: -5.84476 }
  }
]
```

3) Recupera únicamente la población de Vergel.

```
db.cities.find({name: "Vergel"}, {_id: 0, population: 1})
```

```
Atlas atlas-hymc7m-shard-0 [primary] test> db.cities.find({name: "Vergel"}, {_id: 0, population: 1})
[ { population: 4372 } ]
```

4) Calcula el número de ciudades de España.

```
db.cities.find({country: "ES"}).count()
```

```
Atlas atlas-hymc7m-shard-0 [primary] test> db.cities.find({country: "ES"}).count()
2724
```

5) Datos de las ciudades españolas con más de un millón de habitantes.

```
db.cities.find({country: "ES", population: {$gt: 1000000}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] test> db.cities.find({country: "ES", population: {$gt: 1000000}})
[
  {
    _id: ObjectId('69a80347dfae595dec9da080'),
    name: 'Madrid',
    country: 'ES',
    timezone: 'Europe/Madrid',
    population: 3255944,
    location: { latitude: 40.4165, longitude: -3.70256 }
  },
  {
    _id: ObjectId('69a8034bdfae595dec9da2a7'),
    name: 'Barcelona',
    country: 'ES',
    timezone: 'Europe/Madrid',
    population: 1621537,
    location: { longitude: 41.38879, latitude: 2.15899 }
  }
]
```

6) Recupera la cantidad de ciudades que hay entre Andorra (AD) y España.

```
db.cities.find({$or: [{country: "ES"}, {country: "AD"}]}).count()
```

```
Atlas atlas-hymc7m-shard-0 [primary] test> db.cities.find({$or: [{country: "ES"}, {country: "AD"}]}).count()
2733
```

7) Listado con el nombre y la población de las 10 ciudades más pobladas.

```
db.cities.find({}, {name:1,population:1,_id:0}).sort({population:-1})
```

```
Atlas atlas-hymc7m-shard-0 [primary] test> db.cities.find({}, {name:1,population:1,_id:0}).sort({population:-1})
[
  { name: 'Shanghai', population: 14608512 },
  { name: 'Buenos Aires', population: 13076300 },
  { name: 'Mumbai', population: 12691836 },
  { name: 'Mexico City', population: 12294193 },
  { name: 'Karachi', population: 11624219 },
  { name: 'Delhi', population: 10927986 },
  { name: 'Manila', population: 10444527 },
  { name: 'Moscow', population: 10381222 },
  { name: 'Dhaka', population: 10356500 },
  { name: 'Seoul', population: 10349312 },
  { name: 'São Paulo', population: 10021295 },
  { name: 'Lagos', population: 9000000 },
  { name: 'Jakarta', population: 8540121 },
  { name: 'Tokyo', population: 8336599 },
  { name: 'Zhumadian', population: 8263100 },
  { name: 'Kinshasa', population: 7785965 },
  { name: 'Lima', population: 7737002 },
  { name: 'Cairo', population: 7734614 },
  { name: 'London', population: 7556900 },
  { name: 'City of London', population: 7556900 }
]
Type "it" for more
```

8) Nombre de las distintas zonas horarias en España.

```
db.cities.find({country:'ES'}, {timezone:1,_id:0})
```

```
Atlas atlas-hymc7m-shard-0 [primary] test> db.cities.find({country:'ES'}, {timezone:1,_id:0})
[
  { timezone: 'Europe/Madrid' },
  { timezone: 'Europe/Madrid' },
  { timezone: 'Europe/Madrid' },
  { timezone: 'Europe/Madrid' },
  { timezone: 'Europe/Madrid' },
  { timezone: 'Europe/Madrid' },
  { timezone: 'Europe/Madrid' },
  { timezone: 'Europe/Madrid' },
  { timezone: 'Europe/Madrid' },
  { timezone: 'Europe/Madrid' },
  { timezone: 'Europe/Madrid' },
  { timezone: 'Europe/Madrid' },
  { timezone: 'Europe/Madrid' },
  { timezone: 'Europe/Madrid' },
  { timezone: 'Europe/Madrid' },
  { timezone: 'Europe/Madrid' },
  { timezone: 'Europe/Madrid' },
  { timezone: 'Europe/Madrid' },
  { timezone: 'Atlantic/Canary' },
  { timezone: 'Europe/Madrid' },
  { timezone: 'Europe/Madrid' }
]
```

9) Ciudades españolas que su zona horaria no sea Europe/Madrid.

```
db.cities.find({country:'ES',timezone:{$ne:'Europe/Madrid'}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] test> db.cities.find({country:'ES',timezone:{$ne:'Europe/Madrid'}})
[
  {
    _id: ObjectId('69a80345dfae595dec9d98d4'),
    name: 'Yaiza',
    country: 'ES',
    timezone: 'Atlantic/Canary',
    population: 8777,
    location: { latitude: 28.95678, longitude: -13.76535 }
  },
  {
    _id: ObjectId('69a80345dfae595dec9d9916'),
    name: 'Vilaflor',
    country: 'ES',
    timezone: 'Atlantic/Canary',
    population: 1981,
    location: { latitude: 28.15, longitude: -16.63333 }
  },
  {
    _id: ObjectId('69a80345dfae595dec9d9925'),
    name: 'Valsequillo de Gran Canaria',
    country: 'ES',
    timezone: 'Atlantic/Canary',
    population: 8635,
    location: { latitude: 27.98333, longitude: -15.5 }
  },
  {
    _id: ObjectId('69a80345dfae595dec9d9926'),
    name: 'Valleseco',
    country: 'ES',
    timezone: 'Atlantic/Canary',
    population: 4116,
    location: { latitude: 28.05, longitude: -15.58333 }
  },
```

10) Mostrar las ciudades españolas que comiencen por Ben

```
db.cities.find({country:'ES',name:{$regex:'^Ben',$options:''}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] test> db.cities.find({country:'ES',name:{$regex:'^Ben',$options:''}})
[
  {
    _id: ObjectId('69a80346dfae595dec9d9d02'),
    name: 'Benigànim',
    country: 'ES',
    timezone: 'Europe/Madrid',
    population: 5720,
    location: { latitude: 38.95, longitude: -0.43333 }
  },
  {
    _id: ObjectId('69a80346dfae595dec9d9d03'),
    name: 'Benifaió',
    country: 'ES',
    timezone: 'Europe/Madrid',
    population: 12204,
    location: { latitude: 39.28439, longitude: -0.42598 }
  },
  {
    _id: ObjectId('69a80346dfae595dec9d9d04'),
    name: 'Benitachell',
    country: 'ES',
    timezone: 'Europe/Madrid',
    population: 3630,
    location: { longitude: 38.73333, latitude: 0.15 }
  },
  {
    _id: ObjectId('69a80346dfae595dec9d9d05'),
    name: 'Benifairó de les Valls',
    country: 'ES',
    timezone: 'Europe/Madrid',
    population: 1907,
    location: { latitude: 39.73333, longitude: -0.26667 }
  },
```

11) Devolver las ciudades que su zona horaria sea Atlantic/Canary o Africa/Ceuta.

```
db.cities.find({timezone:{$in:['Atlantic/Canary','Africa/Ceuta']}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] test> db.cities.find({timezone:{$in:['Atlantic/Canary','Africa/Ceuta']}})
[
  {
    _id: ObjectId('69a80345dfae595dec9d98d4'),
    name: 'Yaiza',
    country: 'ES',
    timezone: 'Atlantic/Canary',
    population: 8777,
    location: { latitude: 28.95678, longitude: -13.76535 }
  },
  {
    _id: ObjectId('69a80345dfae595dec9d9916'),
    name: 'Vilaflor',
    country: 'ES',
    timezone: 'Atlantic/Canary',
    population: 1981,
    location: { latitude: 28.15, longitude: -16.63333 }
  },
  {
    _id: ObjectId('69a80345dfae595dec9d9925'),
    name: 'Valsequillo de Gran Canaria',
    country: 'ES',
    timezone: 'Atlantic/Canary',
    population: 8635,
    location: { latitude: 27.98333, longitude: -15.5 }
  },
  {
    _id: ObjectId('69a80345dfae595dec9d9926'),
    name: 'Valleseco',
    country: 'ES',
    timezone: 'Atlantic/Canary',
    population: 4116,
    location: { latitude: 28.05, longitude: -15.58333 }
  },
```

12) Obtén el nombre y la población de las tres ciudades europeas más pobladas

```
db.cities.find({timezone:{$regex:'^Europe/'}},
{name:1,population:1,_id:0}).sort({population:-1})
```

```
Atlas atlas-hymc7m-shard-0 [primary] test> db.cities.find({timezone:{$regex:'^Europe/'}}, {name:1,population:1,_id:0}).sort({population:-1})
[
  { name: 'Moscow', population: 10381222 },
  { name: 'London', population: 7556900 },
  { name: 'City of London', population: 7556900 },
  { name: 'Saint Petersburg', population: 4039745 },
  { name: 'Berlin', population: 3426354 },
  { name: 'Madrid', population: 3255944 },
  { name: 'Roma', population: 2563241 },
  { name: 'Paris', population: 2138551 },
  { name: 'București', population: 1877155 },
  { name: 'Minsk', population: 1742124 },
  { name: 'Hamburg', population: 1739117 },
  { name: 'Warsaw', population: 1702139 },
  { name: 'Budapest', population: 1696128 },
  { name: 'Vienna', population: 1691468 },
  { name: 'Barcelona', population: 1621537 },
  { name: 'Milano', population: 1306661 },
  { name: 'Nizhniy Novgorod', population: 1284164 },
  { name: 'Belgrade', population: 1273651 },
  { name: 'München', population: 1260391 },
  { name: 'Stockholm', population: 1253309 }
]
Type "it" for more
```

13) Calcula la cantidad de ciudades españolas cuyas coordenadas de longitud estén comprendidas entre -0.1 y 0.1.

```
db.cities.find({country:'ES','location.longitude':{$gte:-0.1,$lte:0.1}})
```

```
Atlas atlas-hymc7m-shard-0 [primary] test> db.cities.find({country:'ES','location.longitude':{$gte:-0.1,$lte:0.1}})
[
  {
    _id: ObjectId('69a80345dfae595dec9d98fd'),
    name: 'Vila-real',
    country: 'ES',
    timezone: 'Europe/Madrid',
    population: 51205,
    location: { latitude: 39.93333, longitude: -0.1 }
  },
  {
    _id: ObjectId('69a80346dfae595dec9d9a6f'),
    name: 'Orba',
    country: 'ES',
    timezone: 'Europe/Madrid',
    population: 2100,
    location: { latitude: 38.78333, longitude: -0.06667 }
  },
  {
    _id: ObjectId('69a80346dfae595dec9d9b79'),
    name: 'Jalón',
    country: 'ES',
    timezone: 'Europe/Madrid',
    population: 2437,
    location: { latitude: 38.74063, longitude: -0.01129 }
  },
  {
    _id: ObjectId('69a80346dfae595dec9d9c8e'),
    name: 'Castelló de la Plana',
    country: 'ES',
    timezone: 'Europe/Madrid',
    population: 180005,
    location: { latitude: 39.98333, longitude: -0.03333 }
  },
```

14) Mostrar los 3 países con mayor número de ciudades.

```
db.cities.aggregate([{$group: {_id: "$country",cityCount: {$sum: 1}}},{$sort:
{cityCount: -1}},{$limit: 3}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] test> db.cities.aggregate([{$group: {_id: "$country",cityCount: {$sum: 1}}},{$sort: {cityCount: -1}},{$limit: 3}])
[
  { _id: 'FR', cityCount: 8557 },
  { _id: 'DE', cityCount: 6935 },
  { _id: 'IT', cityCount: 6192 }
]
```

15) Devolver los 3 países con mayor número de habitantes.

```
db.cities.aggregate([{$group:{_id:'$country',totalPopulation:{$sum:'$populati
on'}}},{$sort:{totalPopulation:-1}},{$limit:3}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] test> db.cities.aggregate([{$group:{_id:'$country',totalPopulation:{$sum:'$population'}}},{$sort:{totalPopulation:-1}},{$limit:3}])
[
  { _id: 'CN', totalPopulation: 282839031 },
  { _id: 'IN', totalPopulation: 272149640 },
  { _id: 'BR', totalPopulation: 132102085 }
]
```

16) Calcular y devolver la población total de España, Francia e Italia, en orden descendente.

```
db.cities.aggregate([{$match:{country:{$in:['ES','FR','IT']}}},{$group:{_id:'$country',total_population:{$sum:'$population'}}},{$sort:{total_population:-1}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] test> db.cities.aggregate([{$match:{country:{$in:['ES','FR','IT']}}},{$group:{_id:'$country',total_population:{$sum:'$population'}}},{$sort:{total_population:-1}}])
[
  { _id: 'IT', total_population: 56815974 },
  { _id: 'FR', total_population: 51573677 },
  { _id: 'ES', total_population: 44300931 }
]
```

17) Calcular la latitud y longitud media para cada zona horaria europea.

```
db.cities.aggregate([{$match:{timezone:{$regex:'^Europe/'}}},{$group:{_id:'$timezone',avgLatitude:{$avg:'$location.latitude'},avgLongitude:{$avg:'$location.longitude'}}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] test> db.cities.aggregate([{$match:{timezone:{$regex:'^Europe/'}}},{$group:{_id:'$timezone',avgLatitude:{$avg:'$location.latitude'},avgLongitude:{$avg:'$location.longitude'}}}])
[
  {
    _id: 'Europe/Madrid',
    avgLatitude: 34.11325225180677,
    avgLongitude: 3.017071696462533
  },
  {
    _id: 'Europe/Bucharest',
    avgLatitude: 25.027112453038676,
    avgLongitude: 45.803908467771635
  },
  {
    _id: 'Europe/Ljubljana',
    avgLatitude: 14.823260914634147,
    avgLongitude: 46.194177743902436
  },
  {
    _id: 'Europe/Riga',
    avgLatitude: 24.525724186046514,
    avgLongitude: 56.92394255813954
  },
  {
    _id: 'Europe/Helsinki',
    avgLatitude: 24.903318325471698,
    avgLongitude: 62.306457688679245
  },
  {
    _id: 'Europe/Prague',
    avgLatitude: 15.736597498057499,
    avgLongitude: 49.749385866355865
  },
  { _id: 'Europe/Monaco', avgLatitude: 7.42046, avgLongitude: 43.737428 },
  {
    _id: 'Europe/Brussels',
    avgLatitude: 4.565159493433396,
    avgLongitude: 50.746313564727956
  },
```

18) Devolver los países en la posición 5ª y 6ª si ordenamos su población total en orden descendente.

```
db.cities.aggregate([{$group:{_id:'$country',totalPopulation:{$sum:'$population'}}},{$sort:{totalPopulation:-1}},{$skip:4},{$limit:2},{$project:{_id:0,country:'$_id',totalPopulation:1}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] test> db.cities.aggregate([{$group:{_id:'$country',totalPopulation:{$sum:'$population'}}},{$sort:{totalPopulation:-1}},{$skip:4},{$limit:2},{$project:{_id:0,country:'$_id',totalPopulation:1}}])
[
  { totalPopulation: 102632560, country: 'JP' },
  { totalPopulation: 85555230, country: 'MX' }
]
```

19) Calcular las zonas horarias con mayor y menor número de habitantes para España y Rusia. Añade también un campo extra que indique cuántas zonas horarias tiene cada país.

```
db.cities.aggregate([{$match:{country:{$in:['ES','RU']}}},{$group:{_id:{country:'$country',timezone:'$timezone'},totalPopulation:{$sum:'$population'}}},{$group:{_id:'$_id.country',timezones:{$push:{timezone:'$_id.timezone',totalPopulation:'$totalPopulation'}},timezoneCount:{$sum:1}}},{$project:{_id:0,country:'$_id',timezoneCount:1,maxTimezone:{$arrayElemAt:[{$slice:[{$sortArray:{input:'$timezones',sortBy:{totalPopulation:-1}}},1]},0]},minTimezone:{$arrayElemAt:[{$slice:[{$sortArray:{input:'$timezones',sortBy:{totalPopulation:1}}},1]},0]}}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] test> db.cities.aggregate([{$match:{country:{$in:['ES','RU']}}},{$group:{_id:{country:'$country',timezone:'$timezone'},totalPopulation
:{$sum:'$population'}}},{$group:{_id:'$_id.country',timezones:{$push:{timezone:'$_id.timezone',totalPopulation:'$totalPopulation'}},timezoneCount:{$sum:1}}},{$project:{_id
:0,country:'$_id',timezoneCount:1,maxTimezone:{$arrayElemAt:[{$slice:[{$sortArray:{input:'$timezones',sortBy:{totalPopulation:-1}}},1]},0]},minTimezone:{$arrayElemAt:[{$sl
ice:[{$sortArray:{input:'$timezones',sortBy:{totalPopulation:1}}},1]},0]}}}])
[
  {
    timezoneCount: 16,
    country: 'RU',
    maxTimezone: { timezone: 'Europe/Moscow', totalPopulation: 65316206 },
    minTimezone: { timezone: 'Asia/Anadyr', totalPopulation: 30263 }
  },
  {
    timezoneCount: 3,
    country: 'ES',
    maxTimezone: { timezone: 'Europe/Madrid', totalPopulation: 41964016 },
    minTimezone: { timezone: 'Africa/Ceuta', totalPopulation: 152134 }
  }
]
```

20) Calcular qué porcentaje de la población representa la ciudad más poblada de España.

```
db.cities.aggregate([{$match:{country:'ES'}},{$group:{_id:null,totalPopulation:{$sum:'$population'},maxPopulation:{$max:'$population'}}},{$project:{porcentaje:{$multiply:[{$divide:['$maxPopulation','$totalPopulation']},100]}}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] test> db.cities.aggregate([{$match:{country:'ES'}},{$group:{_id:null,totalPopulation:{$sum:'$population'},maxPopulation:{$max:'$popula
tion'}}},{$project:{porcentaje:{$multiply:[{$divide:['$maxPopulation','$totalPopulation']},100]}}}])
[ { _id: null, porcentaje: 7.349696264482341 4 } ]
```

21) Utilizando unwind, devuelve los pares únicos (país y timezone) con un campo que indique la suma total de población del país.

```
db.cities.aggregate([{$group:{_id:'$country',totalPopulation:{$sum:'$population'},timezones:{$addToSet:'$timezone'}}},{$unwind:'$timezones'},{$project:{_id:0,country:'$_id',timezone:'$timezones',totalPopulation:1}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] test> db.cities.aggregate([{$group:{_id:'$country',totalPopulation:{$sum:'$population'},timezones:{$addToSet:'$timezone'}}},{$unwind:
$timezones'},{$project:{_id:0,country:'$_id',timezone:'$timezones',totalPopulation:1}}])
[
  {
    totalPopulation: 2322134,
    country: 'CG',
    timezone: 'Africa/Brazzaville'
  },
  { totalPopulation: 5564244, country: 'IL', timezone: 'Asia/Jerusalem' },
  {
    totalPopulation: 132102085,
    country: 'BR',
    timezone: 'America/Recife'
  },
  {
    totalPopulation: 132102085,
    country: 'BR',
    timezone: 'America/Araguaina'
  },
  {
    totalPopulation: 132102085,
    country: 'BR',
    timezone: 'America/Manaus'
  },
  {
    totalPopulation: 132102085,
    country: 'BR',
    timezone: 'America/Maceio'
  },
```

22) Averigua qué nombres de ciudades se repiten entre distintos países.

```
db.cities.aggregate([{$group:{_id:{name:'$name',country:'$country'},count:{$sum:1}}},{$group:{_id:'$_id.name',countries:{$addToSet:'$_id.country'},numCountries:{$sum:1}}},{$match:{numCountries:{$gt:1}}},{$project:{_id:0,cityName:'$_id',countries:1}}])
```

```
Atlas atlas-hymc7m-shard-0 [primary] test> db.cities.aggregate([{$group:{_id:{name:'$name',country:'$country'},count:{$sum:1}}},{$group:{_id:'$_id.name',countries:{$addToS
et:'$_id.country'},numCountries:{$sum:1}}},{$match:{numCountries:{$gt:1}}},{$project:{_id:0,cityName:'$_id',countries:1}}])
[
  { countries: [ 'ES', 'IT' ], cityName: 'Teulada' },
  { countries: [ 'DE', 'HR' ], cityName: 'Kamen' },
  { countries: [ 'CH', 'DE' ], cityName: 'Burgdorf' },
  { countries: [ 'CO', 'PY' ], cityName: 'Cerrito' },
  { countries: [ 'DE', 'CH' ], cityName: 'Trogen' },
  { countries: [ 'PR', 'PH' ], cityName: 'Santa Barbara' },
  { countries: [ 'AR', 'MX' ], cityName: 'Las Varas' },
  { countries: [ 'RU', 'CO' ], cityName: 'Chita' },
  { countries: [ 'PY', 'AR', 'DO' ], cityName: 'Villa Elisa' },
  {
    countries: [
      'PH', 'AR', 'GT',
      'HN', 'MX', 'CU',
      'CO', 'PE'
    ],
    cityName: 'San Luis'
  },
  { countries: [ 'MX', 'CA' ], cityName: 'Delta' },
  { countries: [ 'IN', 'IT' ], cityName: 'Londa' },
  { countries: [ 'CH', 'DE' ], cityName: 'Laufenburg' },
  {
    countries: [ 'AR', 'EC', 'CU', 'CO', 'MX' ],
    cityName: 'San Cristóbal'
  },
  { countries: [ 'DE', 'CH' ], cityName: 'Waldkirch' },
  { countries: [ 'PH', 'IT' ], cityName: 'Orani' },
  { countries: [ 'CO', 'HN', 'PH' ], cityName: 'Armenia' },
  { countries: [ 'ID', 'PH' ], cityName: 'Pandan' },
  { countries: [ 'CA', 'JM', 'GB', 'NZ' ], cityName: 'Wakefield' },
  { countries: [ 'PT', 'BR' ], cityName: 'Bragança' }
]
```